"Evolutionary dynamics of phenotypically structured populations in time-periodic environments"

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joint work with Sepideh MIRRAHIMI

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Motivations and Questions

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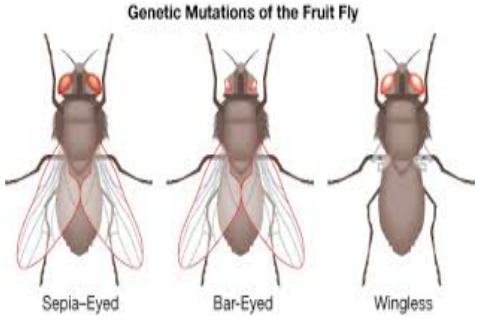

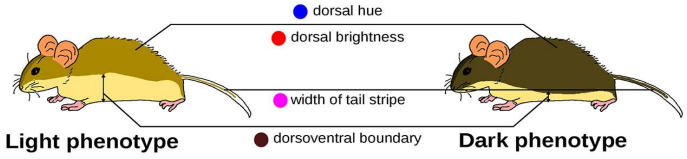

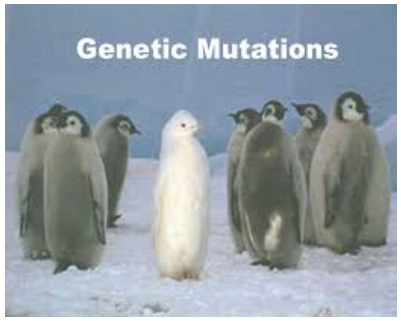

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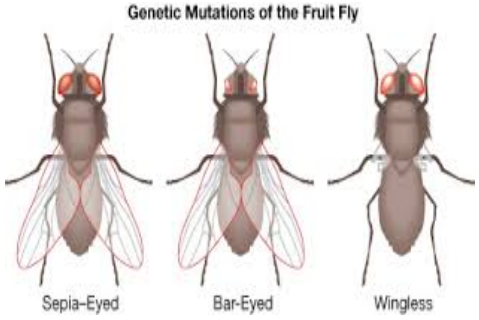

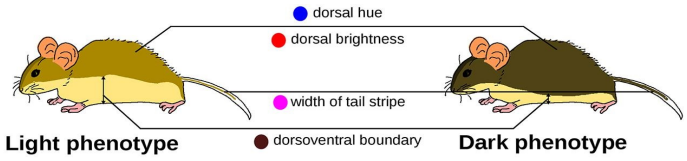

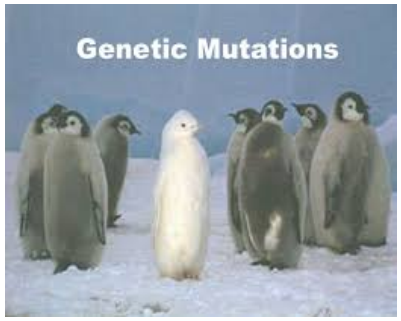

What will be the impact of the variations of the environment on the population's phenotypical distribution?
Periodic variations: seasonal effects, prescription of drugs, ...

Biological experience (Ketola et al. 2013)

Evolution of a population of bacterial pathogen *Serratia marcescens*. Several populations of bacteria kept in environments with constant or fluctuating temperature for several weeks and measured their growth rates in different environments.

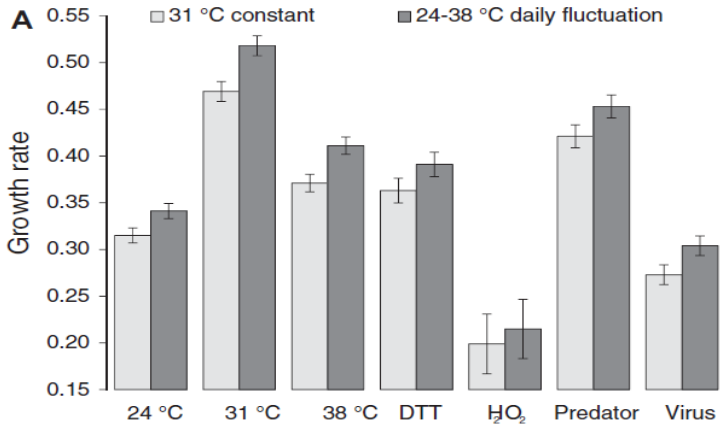

Figure: Average maximal growth rate of Serratia marcescens-bacteria evolved either in constant $31°$C or in daily fluctuating temperature between $24°$C and $38°$C, (Figure from Ketola et al. 2013)

Model and Notation

A parabolic integro-differential model with $(t,x) \in [0,+\infty) \times \mathbb{R}^d$

$$\begin{cases} \partial_t n(t,x) - \sigma \Delta n(t,x) = n(t,x)[a(t,x) - \rho(t)], \\ \rho(t) = \displaystyle\int_{\mathbb{R}^d} n(t,x)dx, \\ n(t=0,x) = n_0(x). \end{cases} \tag{P}$$

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- $n(t,x)$: density of individuals with trait x at time t.
- $\sigma > 0$: rate of mutations.
- $a(t,x)$: net growth rate of individuals.
- $\rho(t)$: death term due to competition.

Assumptions

The function $a(t,x) \in C^3$ in x is $T-$periodic in t such that

$$\|a\|_{L^\infty} \leq d_0,$$

$$\exists!\ x_m : \overline{a}(x_m) = \max_{x \in \mathbb{R}^d} \overline{a}(x) > 0, \qquad \overline{a}(x) = \frac{1}{T}\int_0^T a(t,x)dt.$$

Also,

$$0 \leq n_0(x) \leq e^{C_1 - C_2|x|}, \quad \forall x \in \mathbb{R}^d.$$

Previous Works

Hamilton-Jacobi Approach in evolutionary dynamics of populations

- O.DIEKMANN, P.E.JABIN, S.MISCHLER, B.PERTHAME, *The dynamics of adaptation: an illuminating example and a Hamilton-Jacobi approach*, (2005).
- B.PERTHAME, G.BARLES, *Dirac concentrations in Lotka-Volterra parabolic PDEs*, (2008).
- (...) *widely extended to more complex models*.

Time-varying environment

- S.MIRRAHIMI, B.PERTHAME AND P.SOUGANIDIS, *Time fluctuations in a population model of adaptive dynamics*, (2015). (*different scaling*).

Summary

1 Introduction and Motivation

2 Main results

3 Some Heuristics and the Approximation of Moments

4 Biological Examples

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Convergence to a periodic solution

Proposition

If a is "small" outside of a compact and $\sigma << 1$, then

$$|\rho(t) - \overline{\rho}(t)| \to 0,$$

$$\|n(t,\cdot) - \overline{n}(t,\cdot)\|_{L^\infty} \to 0, \quad \textit{as } t \to \infty,$$

where $(\overline{n}, \overline{\rho})$ is the unique periodic solution of (P).

Periodic Model

Let $\sigma = \varepsilon^2$, we study $(n_\varepsilon, \rho_\varepsilon)$, the unique periodic solution of the equation:

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Hopf-Cole transformation

$$n_\varepsilon = \frac{1}{\sqrt{2\pi\varepsilon}} \exp\left(\frac{u_\varepsilon}{\varepsilon}\right).$$

Case $\sigma = \varepsilon^2$, asymptotic behavior

Theorem

(i) *As* $\varepsilon \to 0$*:*

$$\|\rho_\varepsilon(t) - \widetilde{\varrho}(t)\|_{L^\infty} \to 0, \quad \text{and} \quad n_\varepsilon(t,x) - \widetilde{\varrho}(t)\delta(x - x_m) \rightharpoonup 0,$$

where $\widetilde{\varrho}(t)$ *is periodic.*

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where $\widetilde{\varrho}(t)$ *is periodic.*

(ii) *Moreover as* $\varepsilon \to 0$*,* u_ε *converges to* $u(x) \in C^3(\mathbb{R}^d)$*, the unique viscosity solution of:*

$$\begin{cases} -|\nabla u|^2 = \overline{a}(x) - \overline{a}(x_m), & x \in \mathbb{R}^d, \\ \max\limits_{x \in \mathbb{R}^d} u(x) = u(x_m) = 0, \end{cases}$$

which can be explicitly computed.

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Heuristics

Let $u_\varepsilon = \varepsilon \ln n_\varepsilon$, then for $(t,x) \in [0,+\infty) \times \mathbb{R}^d$:

$$\frac{1}{\varepsilon}\partial_t u_\varepsilon - \varepsilon \Delta u_\varepsilon = |\nabla u_\varepsilon|^2 + a(t,x) - \rho_\varepsilon(t).$$

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Formally asymptotic expansion for u_ε with T-periodic coefficients.:

$$u_\varepsilon(t,x) = u(t,x) + \varepsilon v(t,x) + \varepsilon^2 w(t,x) + o(\varepsilon^2),$$

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$$n_\varepsilon(t,x) \approx \frac{1}{\sqrt{2\pi\varepsilon}} e^{\frac{u(x)}{\varepsilon} + v(t,x) + \varepsilon w(t,x)}.$$

Approximation of distribution's moments

Equation for $u(x)$:

$$u(x) = -\left| \int_{x_m}^{x} \sqrt{\overline{a}(x_m) - \overline{a}(y)}\, dy \right|,$$

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$$\begin{cases} \partial_t v &= a(t,x) - \overline{a}(x) - \rho(t) + \overline{\rho}, \\ -\Delta u &= \dfrac{2\nabla u}{T} \displaystyle\int_0^T \nabla v dt - \Delta u(x_m). \end{cases}$$

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Since $u(x) < 0$ where $x \neq x_m$ we have $n_\varepsilon << 1$ and we use the Laplace Method to approximate the distribution's moments of the population.

Notations for distribution's moments

Mean Total population: $\overline{\rho} = \int_0^T \rho_\varepsilon(t)dt = \int_0^T \int_{\mathbb{R}^d} n_\varepsilon(t,x)dxdt$

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Mean Fitness: $F(\tau) = \int_{\mathbb{R}^d} a(\tau,x)\dfrac{1}{T}\int_0^T \dfrac{n_\varepsilon(t,x)}{\rho_\varepsilon(t)}dtdx$

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$$\mu_p(t) \approx \frac{2\varepsilon c}{b}\sqrt{g}\sin\left(bt - \frac{\pi}{2}\right), \quad \sigma_p^2 \approx \frac{\varepsilon}{\sqrt{g}}, \quad \overline{\rho}_\varepsilon \approx r - \frac{gc^2}{2} - \varepsilon\sqrt{g},$$

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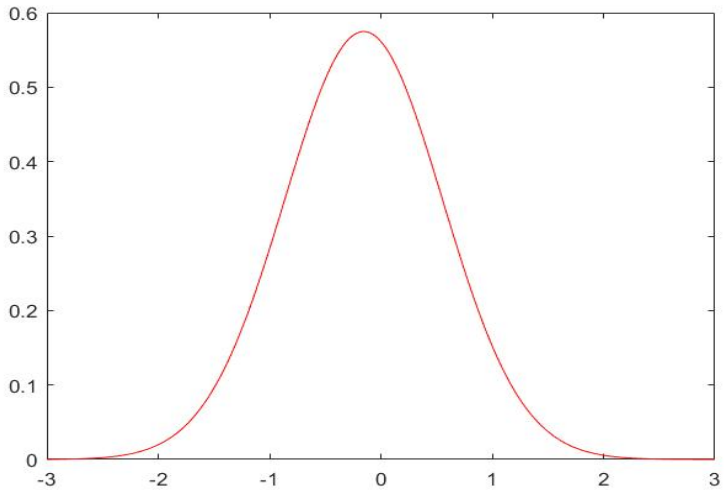
- Moments for population evolved in constant environment ($t = 1/2$):

$$\mu_c = 0, \quad \sigma_c^2 = \frac{\varepsilon}{\sqrt{g(1/2)}}, \quad \overline{\rho}_\varepsilon = r - \varepsilon\sqrt{g}, \quad F_c = r - \varepsilon\sqrt{g}$$

Convergence to Dirac mass

$$a(t,x) = r - g(x - c\sin bt)^2$$

$r = 2,\ g = c = 1,\ b = 2\pi.$
$\varepsilon = 0.5$



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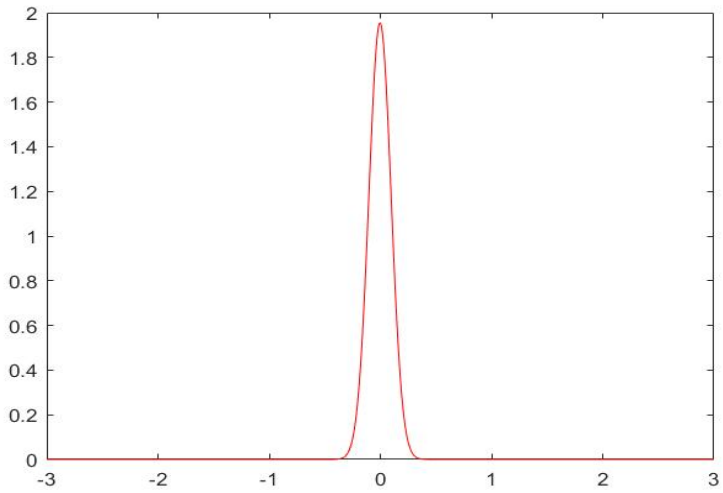

Figure: Concentration about the optimal trait for different mutation rates.

Load in the phenotypical mean

$$a(t,x) = r - g(x - c\sin bt)^2 \qquad \mu_p(t) \approx \frac{2\varepsilon c}{b}\sqrt{g}\sin\left(bt - \frac{\pi}{2}\right)$$

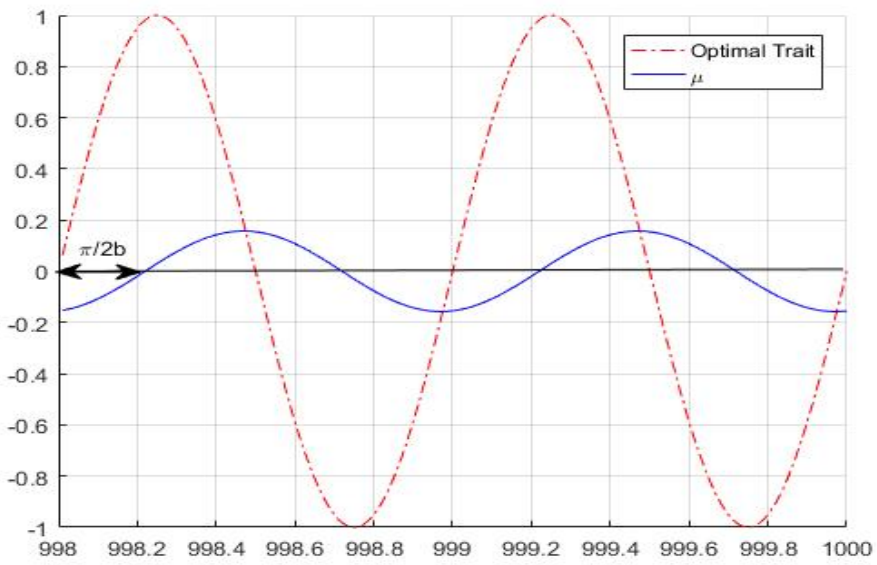

Figure: Comparison between the optimal trait and the phenotipycal mean (delay=1/4).

Oscillations on the pressure of the selection

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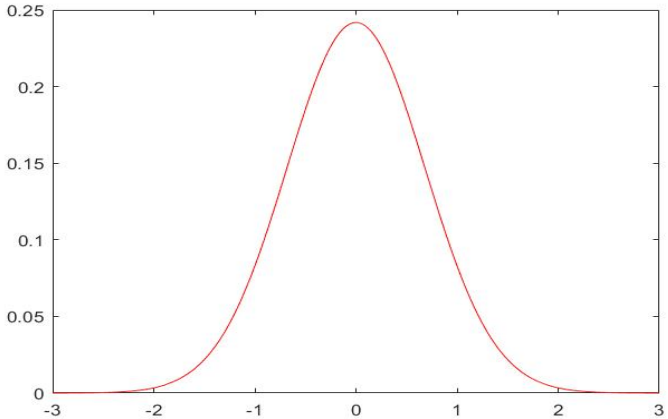

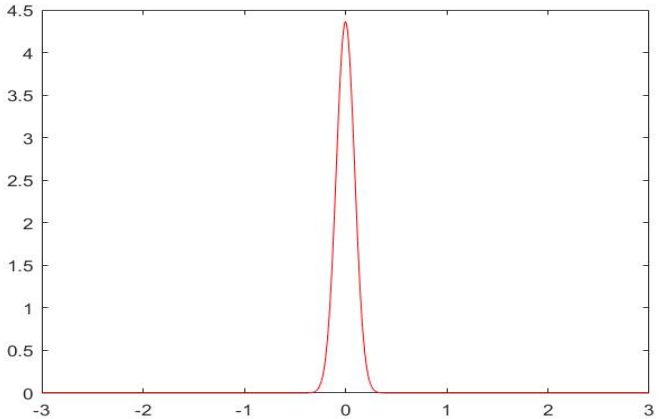

Figure: Concentration about the optimal trait for different mutation rates.

Other numerical examples

Case for "two symmetric maxima"

$$a(t,x) = \begin{cases} (x-x_1)^2(x-x_2)^2(1-x), & \text{if } t \in (0,1/2), \\ (x-x_1)^2(x-x_2)^2 x, & \text{if } t \in (1/2,1) \end{cases}$$

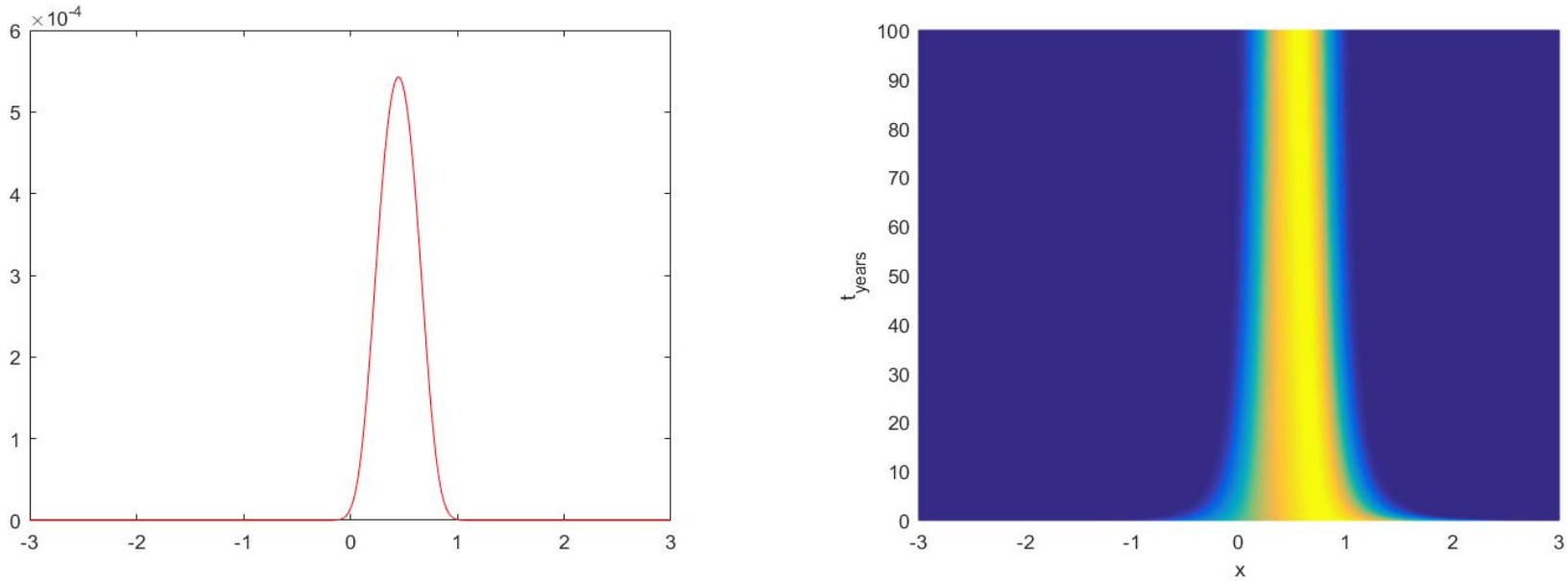

Figure: Concentration about the optimal trait, left: at end time, right: for all values of t. $(x_1 = 0.4, x_2 = 0.6)$

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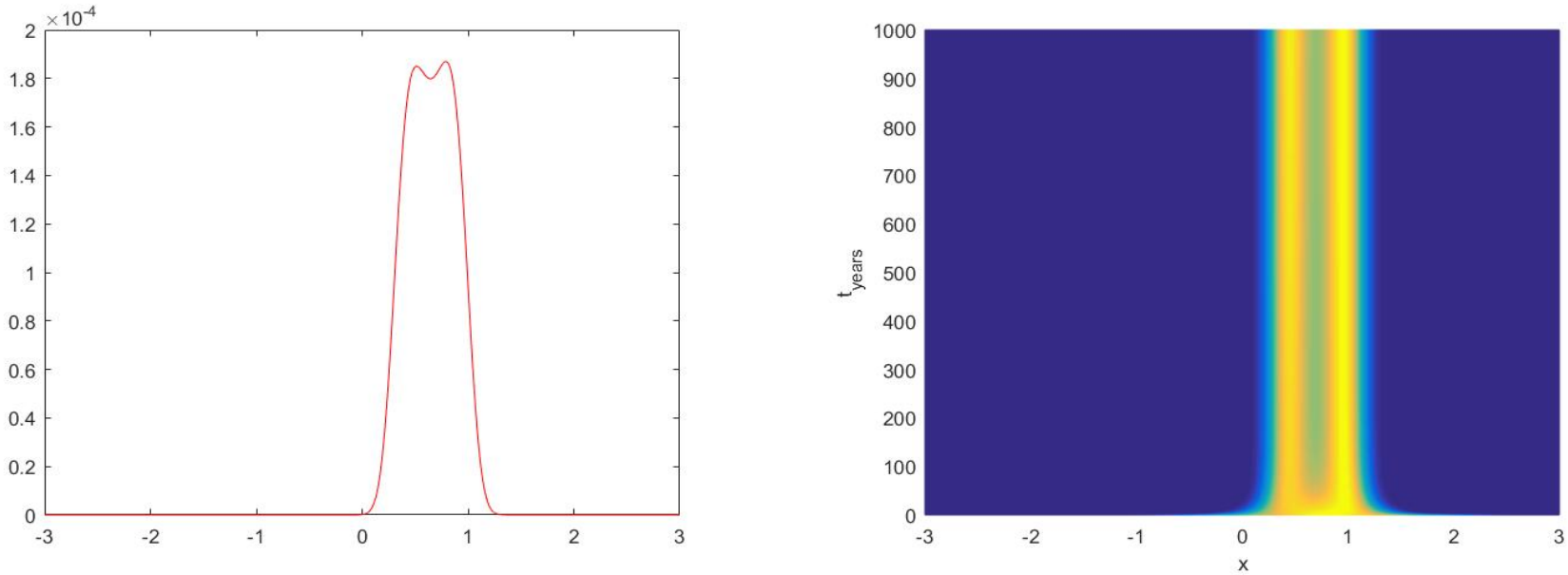

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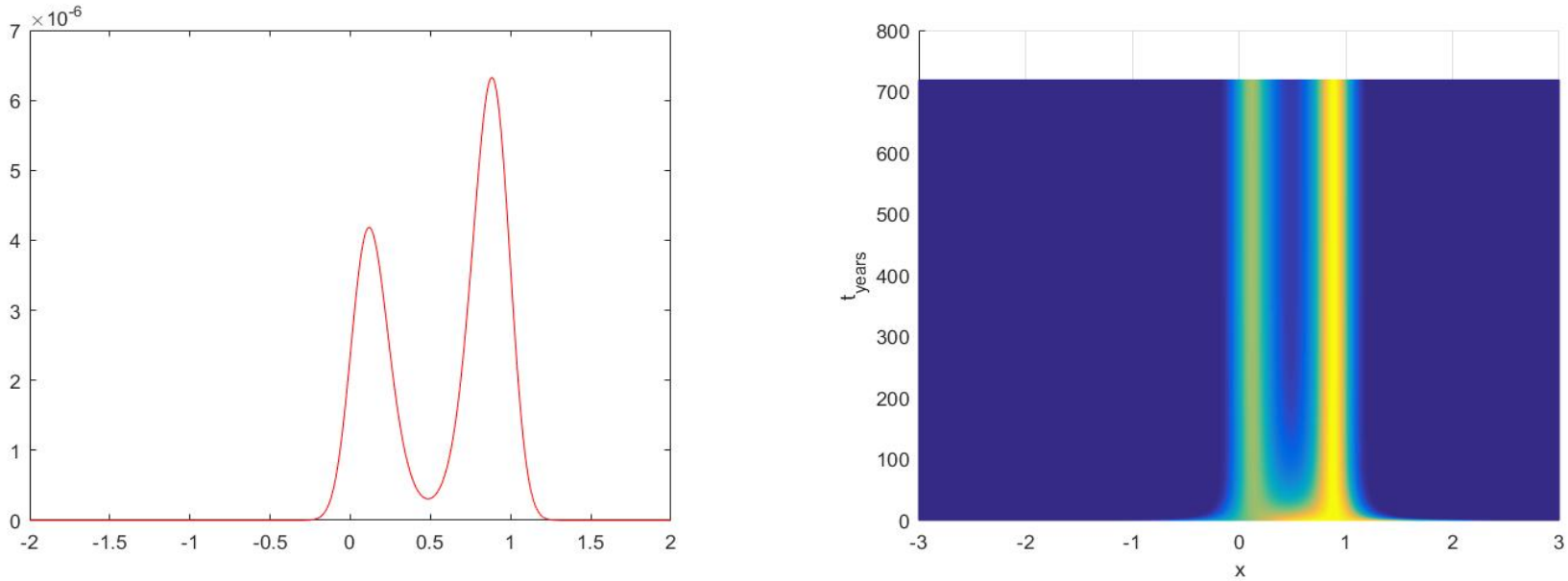
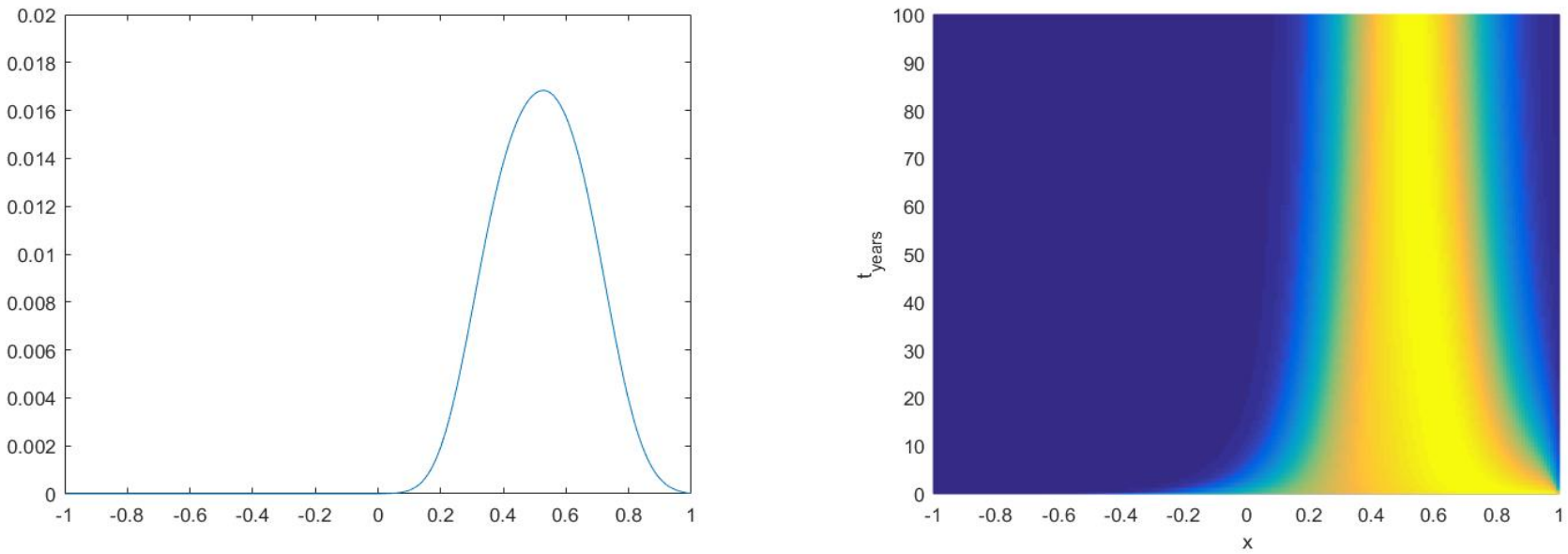


Figure: Concentration about the optimal trait, left: at end time, right: for all values of t. $(x_1 = 0.1, x_2 = 0.9)$

Other numerical examples

Case for "two non-symmetric maxima"

$$a(t,x) = \begin{cases} R(x)(1-x), & \text{if } t \in (0,1/2), \\ R(x)x, & \text{if } t \in (1/2,1). \end{cases} \quad R(x) = -(x-3)^2(x-\frac{1}{2})^4$$



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Conclusions and future work

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- We obtain good approximations for the distribution's moments of the population.

$\rightarrow$ Study of the impact of a climate shift: $a(t, x - ct)$ (work in progress).

THANK YOU!

MERCI!

¡GRACIAS!